



Makasna

SRT GATEWAY PLATFORM

LIVE VIDEO TRANSPORT. BUILT FOR OPERATIONS.

Receive. Monitor. Protect. Route. Deliver.

Self-hosted live contribution gateway with multi-destination fan-out, source failover evaluation, and sub-second operational telemetry.

Self-hosted gateway

- SRT contribution
- Multi-destination routing
- Source and destination health
- IP-to-SDI integration
- Recovery and failover evaluation

CONTENTS

A clearer way to operate contribution.

What follows: the workflow, the controls, the interfaces and a practical evaluation path.

Reader promise

Clear buyer story first.

Technical evidence next.

Qualification points stay visible so your engineering team can evaluate fit with mission-critical confidence.

01 Why Makasna

02 Architecture and workflow

03 Use cases

04 Route operations

05 Monitoring

06 Interfaces and SRT

07 Recovery and failover

08 Deployment and PoC

WHY MAKASNA

Live video is a workflow problem.

Connections are only the beginning. Operators need context when a feed changes, a receiver drops or a source fails.

Unstable networks

Packet loss, fluctuating latency and changing reachability affect contribution paths unpredictably across public transit networks.

Multiple outputs

One source feed often needs production, backup, monitoring and local facility infrastructure at the same time.

Limited context

A single connection indicator alone does not show whether the source encoder, the network route, or the downstream receiver is at fault.

Manual recovery

Restarting separate ad-hoc processes under live-event pressure creates avoidable operational and commercial risk.

Treat transport as workflow: define routes, inspect each side, recover from evidence.

THE SOLUTION

One route. One operating view.

Makasna places source configuration, destination fan-out, transport health and lifecycle controls in one workspace.

Receive

SRT and supported IP sources enter through a configured route endpoint.

Operate

Operators inspect controls, status, real-time statistics and recent events together.

Deliver

SRT, UDP and qualified DeckLink SDI outputs serve downstream production systems.

The route is the unit of operation. It holds the source, destinations, policy and state needed for a repeatable live workflow.

Control plane

Supervision loop, REST API, WebSocket stats, event logging

Media plane

Native C / GStreamer / FFmpeg engine, SRT/UDP sinks, DeckLink SDI

Persistent state

Configuration store, automatic backups, license state and route recovery

ARCHITECTURE

A gateway between contribution and delivery.

The deployed system separates the control plane supervision from high-performance native media engines.

Web operator

- Browser UI & REST API
- Route controls & configuration
- Live statistics & charts
- Audit log inspection

Control plane

- Process lifecycle supervision
- Failover policy engine
- Transactional state store
- Route watchdog & reconnect

Native engine

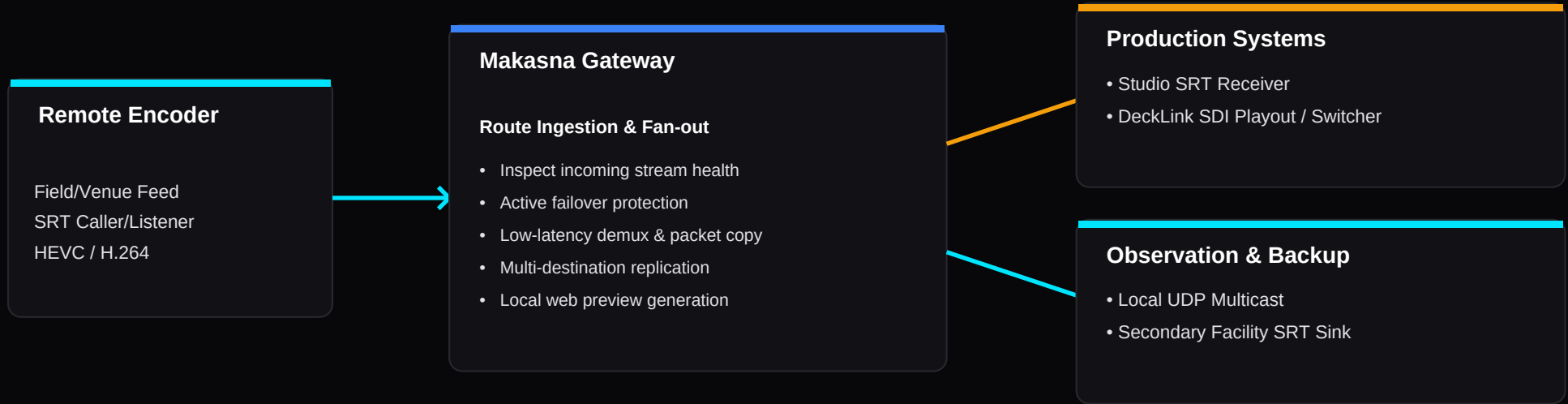
- GStreamer & MediaMTX pipeline
- SRT caller/listener sockets
- DeckLink SDI playout
- Low-latency stream copy

Control and media responsibilities stay separate: the control plane supervises route state while the native engine handles stream processing.

WORKFLOW

From encoder to production path.

Receive once at the gateway, then create the delivery paths the production workflow requires.



The gateway adds delivery destinations while keeping each network address, port, mode and format explicit.

USE CASE 01

Venue contribution for live events.

Keep the source workflow simple while the gateway distributes the contribution feed to production and monitoring.

Source site

One single hardware encoder sends one contribution path from the venue.

Production

Main production control room consumes the primary feed via low-latency SRT.

Backup

Secondary receiver or disaster recovery facility stays warm and available.

Monitoring

Local observation via UDP or HLS browser preview without adding load to encoder.

Best fit when one contribution source feeds production, backup and monitoring from a centralized gateway.

USE CASE 02

Remote production and studio distribution.

Organize recurring sources and outputs around events, venues and operating roles.

Organize

Name routes by event, venue and role so operators instantly understand feed context.

Repeat

Clone recurring route shapes with one click, then review port endpoints.

Observe

Inspect source and destination independently without disrupting ongoing live video.

Handover

Record active routes and approved configuration changes in an immutable event log.

Best fit when recurring broadcast productions require standardized route nomenclature, repeatable roles and clean shifts.

USE CASE 03

IP contribution into SDI infrastructure.

Bridge supported IP contribution into compatible Blackmagic DeckLink workflows.

Input

Supported contribution enters through SRT or UDP IP network streams.

Decode

Native GStreamer pipeline handles hardware-accelerated media decoding and conversion.

Playout

DeckLink output targets baseband broadcast infrastructure with genlock capability.

Qualify

Test video mode, audio channels, and sustained lock before live production rollout.

Test target hardware, PCIe drivers, source video formats, and embedded multi-channel audio before air.

ROUTE OPERATIONS

Configure. Check. Start.

A repeatable operator workflow reduces setup ambiguity and makes operational handover seamless.

Route list & workspace

NAME	ENABLED	INPUT	OUTPUT	ACTIONS
 Venue contribution - local demo	YES	SRT:listener:12100	UDP:127.0.0.1:12200, SRT:12300	 Stop  []
 Studio contribution - standby demo	NO	SRT:listener:12102	SRT:12302 (Backup Facility)	 Start  []

Synthetic route evidence; names and endpoints are lab-configured to demonstrate non-disruptive fan-out.

ROUTE OPERATIONS

Manage the route through its lifecycle.

Start, stop, restart, clone and edit with clear awareness of output interruption.

Operational Lifecycle Matrix

ACTION	OPERATIONAL MEANING	CHECK AFTER ACTION
Start	Launch route and media process	Source state, receiver playback
Stop	End route and downstream delivery	Receiver behavior, event entry in log
Restart	Recreate route media pipeline	Recovery time, output interruption interval
Edit / clone	Change or duplicate configuration	Ports, credentials, downstream fan-out endpoints

Source or destination edits can restart a running route. Plan configuration updates around the interruption tolerance of the live workflow.

FAN-OUT

One contribution. Multiple destinations.

Add the outputs required by production, monitoring, backup and downstream distribution systems.

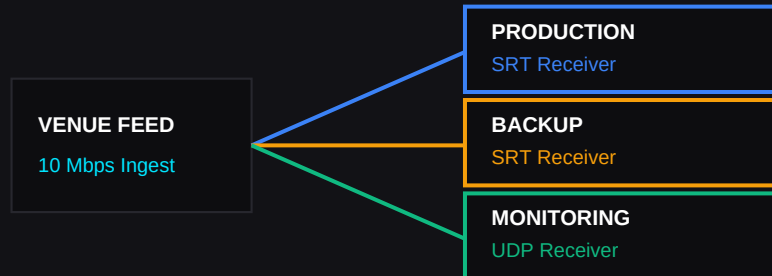
Bandwidth Arithmetic

Illustrative bandwidth arithmetic:

A 10 Mbps incoming feed replicated to 3 receivers requires approximately 30 Mbps of payload egress, plus SRT protocol overhead and ARQ recovery headroom.

Direct stream copy uses 0% CPU.

Topology Fan-out



Confirm actual network egress rates and ISP uplink headroom during technical qualification.

MONITORING

See the transport. Verify the delivery.

Live telemetry helps operators distinguish source, gateway and destination symptoms.

Real-time Telemetry Surface

Healthy source (Primary)

Receive rate: ↓ 4.85 Mbps
RTT Latency: 12.4 ms
Packet Loss: 0.00% (0 packets)
Resolution: 1920x1080 @ 50 fps

Destination Check

Destination 1: SRT Caller (12300) · 1 client
Egress Rate: ↑ 4.88 Mbps
Destination 2: UDP Multicast (12200)
Delivery State: Sustained Lock / Active

Inspect outgoing client state alongside actual receiver playback to confirm complete pipeline health.

MONITORING

Measurements need operational context.

Use rate, RTT, loss, frame information and client state alongside actual receiver playback.

Receive rate

Compare incoming data rate with expected source bitrate to catch encoder throttling.

RTT and loss

Look for sudden network path changes, peer buffer mismatch, or route congestion.

Resolution / FPS

Compare probed video metadata with agreed broadcast delivery format specifications.

Client state

Check destination connection status and playback together to rule out edge drops.

RTT and link bandwidth are transport observations, not end-to-end latency or certified capacity.

INTERFACES

Connect the right endpoints.

Protocol support is the starting point; formats, ports, settings and dependencies complete the path.

Protocol & Interface Matrix

DIRECTION	INTERFACE	QUALIFICATION FOCUS
Input	SRT / UDP	Peer settings, reachable UDP ports, payload media format
Input	RTMP / HLS / HTTP-FLV	FFmpeg remux path, ingestion URL access, authentication tokens
Output	SRT / UDP	Egress uplink capacity, downstream receiver compatibility, latency buffer
Output	DeckLink SDI	PCIe card compatibility, drivers, GStreamer plugins, audio channel map

Validate end-to-end format negotiation and clock alignment before deploying live multi-channel production.

SRT

Reliable transport with explicit settings.

Agree connection mode, latency buffer, Stream ID and passphrase-based AES encryption with the peer.

Connection mode

Caller, listener or rendezvous must match the network topology and NAT/firewall constraints.

Latency buffer

Choose appropriate RTT multiplier (typically 2.5x to 4x RTT) and test on representative network transit.

Protection

Optional AES-128/256 passphrase encryption; management access network is completely separate.

Align Stream ID, IP address, port, latency buffer, encryption passphrase and key length with the remote peer.

RECOVERY

Plan for the interruption.

Reconnect handling, supervision and watchdog behavior reduce manual recovery steps; they do not promise hitless output.

Supervision

Per-route process supervision isolates native engine faults without taking down adjacent active streams.

Reconnect

Supported contribution paths automatically retry socket connection after temporary IP network outages.

Watchdog

Stalled media pipelines are detected via heartbeat and restarted automatically according to route policy.

Automated recovery reduces manual operator interventions; host, power, network, and source failures still need clear procedures.

FAILOVER

Primary and secondary sources.

Current build supports evaluation of policy-driven source switching and switchback behavior.

Source Failover Policies

POLICY	OPERATIONAL BEHAVIOR
Maintain primary	Prefers primary source; moves to secondary on signal loss and returns when primary stabilizes.
Maintain stability	Stays on current active source until it fails; does not switch back solely because primary recovers.
Manual switchback	Automatic failover to secondary on failure; operator manually returns to primary when ready.
Manual selection	Operator retains complete manual control over active contribution source.

Primary/secondary SRT failover is available for evaluation. Seamless frame-accurate SDI switching is hardware-dependent.

DEPLOYMENT

Run within your infrastructure.

Docker, bare-metal Linux host and virtual-machine deployments each need network, resource and hardware qualification.

Deployment Models

MODEL	ENGINEERING CONSIDERATIONS & RESOURCE PLANNING
Linux Bare-Metal	Dedicated CPU cores, RAM, 10GbE network interfaces, concurrent route stream mix
Docker Container	Reachable UDP ports mapped with host networking, persistent volume mounts for database
Virtual Machine	Reserved vCPU resources, virtio-net paravirtualized network, PCIe passthrough if using SDI
DeckLink Hardware	Compatible Blackmagic PCIe card, Desktop Video driver version, GStreamer plugin build

Separate relay-only routes from decode and SDI playout workloads. Single stream demos cannot extrapolate peak channel density.

SECURITY

Protect the path and the operation.

Plan management access, credentials, configuration backups, licensing connectivity and network boundaries.

Management

Restrict web browser and REST API access strictly to internal management VLAN with reverse proxy HTTPS.

Credentials

Standard administrator account model with cryptographic session cookies and encrypted credential stores.

Configuration

Persist state in transactional database, control configuration exports and test restoration off-line.

Licensing

Activation and periodic lease validation operate through private license service with offline node support.

Enforce strict network segmentation between live video transport multicast/SRT networks and administrative control planes.

POC

Turn a demonstration into evidence.

A scoped proof of concept measures the workflow on representative sources, receivers and target infrastructure.

Proof of Concept Evaluation Framework

PHASE	EVIDENCE & QUALIFICATION OBJECTIVES
1. Define	Identify sources, receivers, video codecs, route density, network jitter, and latency thresholds
2. Connect	Validate picture integrity, multi-channel audio, Stream ID matching, and every destination sink
3. Measure	Record RTT, packet loss, bandwidth overhead, CPU/RAM utilization, and hardware thermals
4. Interrupt	Simulate packet drops, pull network cables, test reconnect behavior and source failover policies
5. Operate	Evaluate operator web UI, route cloning, lifecycle actions, backup restoration, and soak testing
6. Review	Synthesize measured test data, document operational boundaries, and agree deployment scope

Structured PoC testing establishes factual engineering baselines before committing to broad operational transition.

NEXT STEP

Bring one real contribution path.

Start with a technical demonstration, agree acceptance criteria and use the results to shape deployment.

Focused Proof-of-Concept Initiation

We will map your source feeds, gateway architecture, and downstream delivery requirements.

1. Share one representative venue contribution feed (SRT or UDP).
2. Define required production, backup, and observation endpoints.
3. Test failover behavior, latency buffers, and network resilience in your real environment.
4. Use measured operational telemetry to validate readiness for mission-critical broadcasts.

Ready to evaluate? Connect directly with the Makasna engineering team to begin your scoped technical PoC.